

Computational biology

Sequence-structure-function paradigm

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Grenoble

Statistiques pour les sciences du Vivant et de l'Homme

September 22, 2026

Goal

- Get an overview of computational biology topics
 - Topics (genomics, metagenomics, proteomics, etc.)
 - Know basic elements in biology (gene to function)
 - Know some important databases
 - Know standard tools (Blast) and libraries (BioPython)
- Have a basic culture of order of magnitude in computational biology
 - Quantity of data
 - Size of genomes
 - Size of organisms
- Toward autonomy for design and implementation of methods
 - Case study of SNP detection
 - Algorithmics on sequences

Lecture organization

- Part I: Genomics
 - Week I: some background in biology, starting your project
 - Week II: Practicals only
 - Week III: Homology and function prediction
- Part II: Structure prediction
 - Week I: history and state-of-the-art in protein structure prediction
 - Week II: Practicals
 - Week III: AI in protein biology & tutorial on AI

project I to be handed-out on the 16th of October.

project II to be handed out on the 6th of November.

Evaluation: project-based + quizz + bonus for participation

Elements of biology

Why studying biology?

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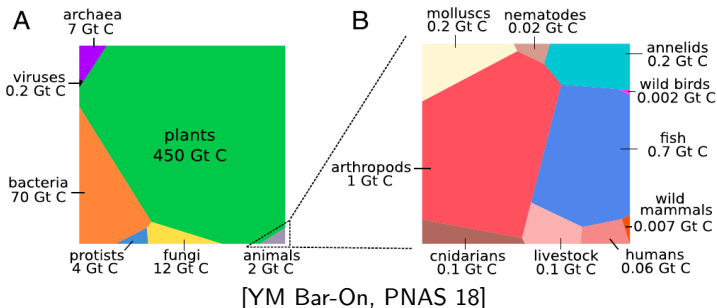
- Just as other sciences: understand the world around us
- For human health: diseases, epidemics, etc.
- For biotech production (e.g. synthesis of materials)
- But also for studying environment

Orders of magnitude: mass repartition

Biology is hardly about humans.

Orders of magnitude: mass repartition

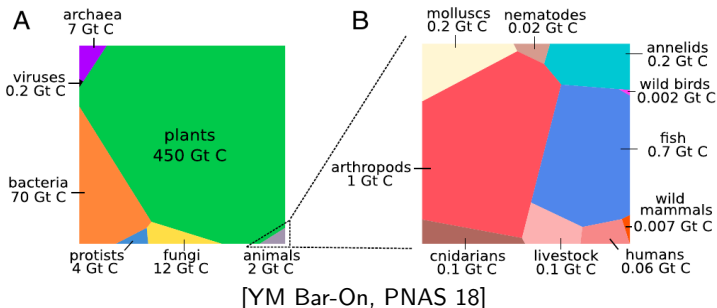
Biology is hardly about humans.



humans $\approx 0.01\%$ global living biomass.

Orders of magnitude: mass repartition

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humans $\approx 0.01\%$ global living biomass.

In term of number of entities and biodiversity, microbes are by far the winners.

Tree of life

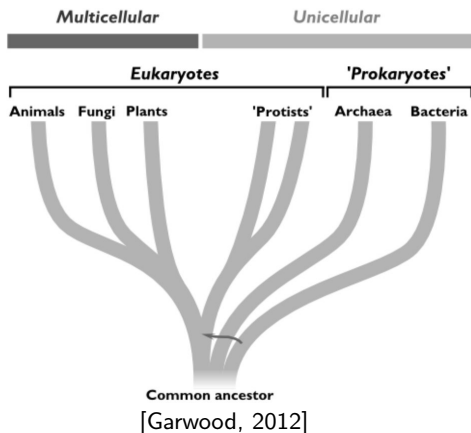
"Nothing in biology makes sense

Tree of life

"Nothing in biology makes sense except in the light of Evolution" T. Dobzhansky

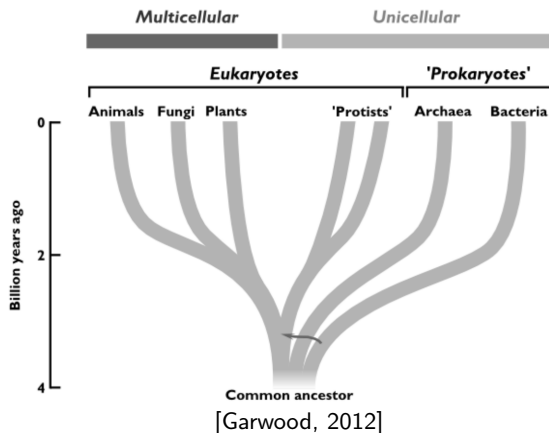
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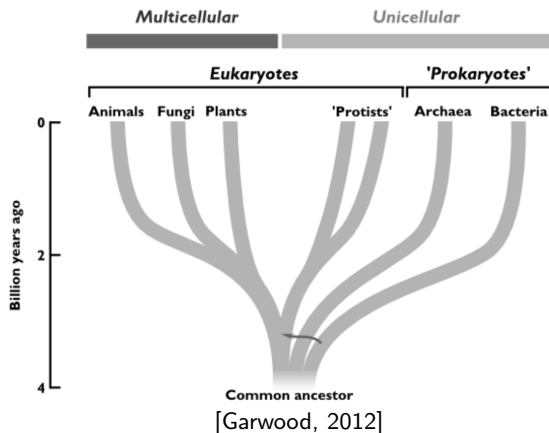
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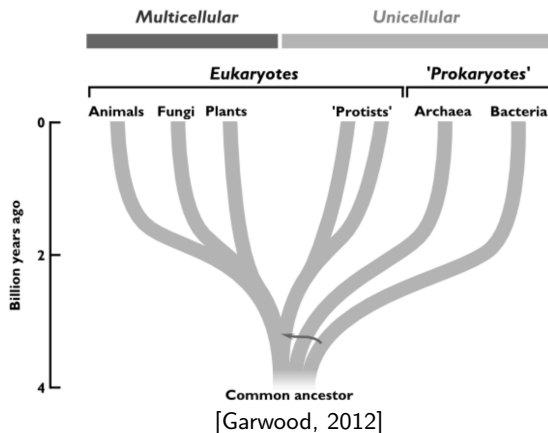
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When was the split between *Homo* and apes?

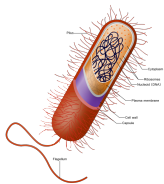
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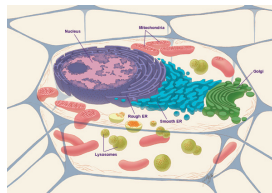


When was the split between *Homo* and apes? $\approx 3\text{M y. ago.}$

Main split: prokaryotes and eukaryotes

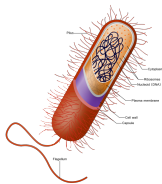


Prokaryotes
"Simple", no nucleus

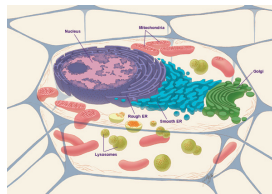


Eukaryotes
Advanced, nucleus

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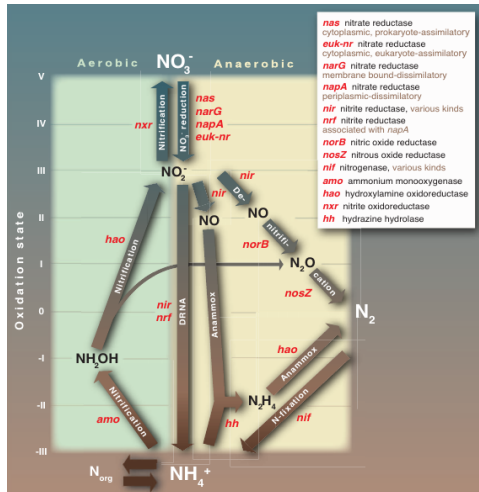


Eukaryotes
Advanced, nucleus

Implication at different stages: cellular mechanisms but also genome structure.

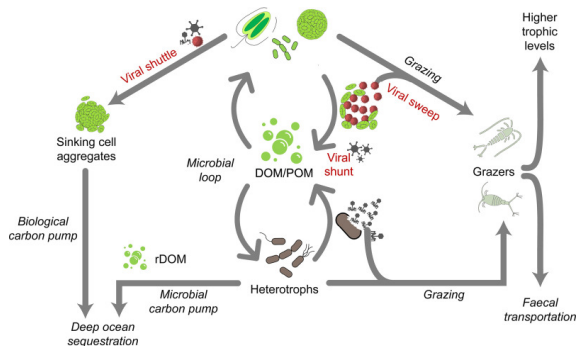
Focus on the microbial world

Prokaryotes make nitrogen available for plants



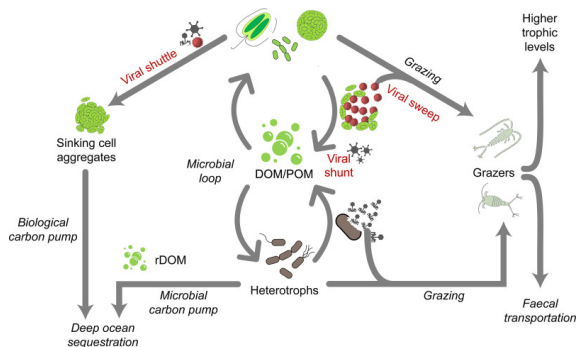
[Canfield et al., Science 2010]

Ocean microbiome pumps CO₂



[Mayers et al., mBio 2023]

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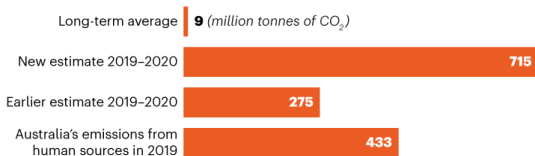
CO₂ turnover: viruses kill 20% of the living biomass in the ocean every day! [Suttle, Nat. Microbiol. 2007]

Microbiomes at swift feedbacks



RECORD EMISSIONS

Devastating fires in southeastern Australia in the summer of 2019–2020 released almost 80 times as much carbon dioxide into the atmosphere as a typical summer bush-fire season.



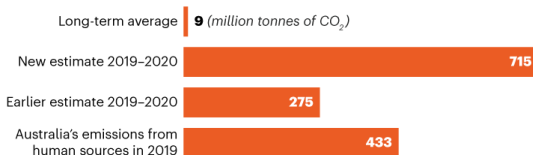
©nature

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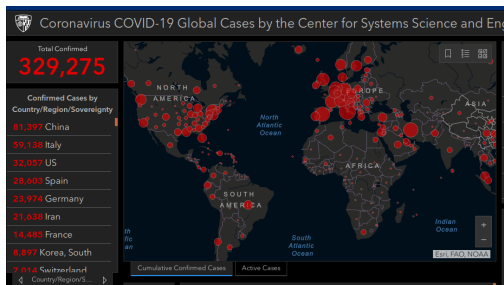
©nature

95% of emitted CO₂ has been pumped down by planktonic bloom.

[Nature 597, 459–460 (2021), Tang et al. Nature (2021)]

Microbes are strongly implied in human health

“Negative” effects:



Covid-19

“Positive” effects:



Health status highly correlated with the diversity of the gut microbiome [Valdes et al. 2018]

The human gut microbiome

2000's
Human genome



$\approx$ 20k protein-coding genes

2010's
Gut metagenomes



The human gut microbiome

2000's
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2010's
Gut metagenomes



$\approx 20\text{k}$ protein-coding genes $\xrightarrow{\times 100}$ $\approx 2\text{M}$ protein-coding genes

Human gut microbiome is rich! What microbes do there is absolutely necessary to keep us alive!

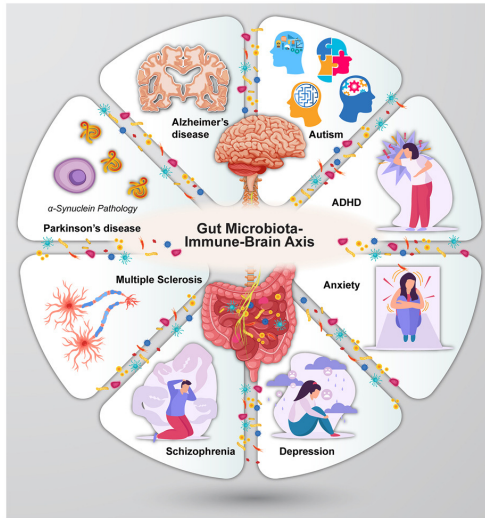
Bacteria-mediated gut metabolic processes

Intestinal metabolic process	Role in human health	Associated pathways	Impacting bacteria	Mode of bacterial modulation	Health impact
Fermentation of dietary fibers ^{188,189}	Produces SCFAs for colonocyte energy and systemic health	GPR41, GPR43 signaling	<i>Faecalibacterium</i> , <i>Roseburia</i>	Enhance SCFA production	Supports gut health, reduces inflammation
Conversion of primary to secondary bile acids ^{199,201}	Aids in lipid digestion and vitamin absorption	FXR, TGR5 signaling	<i>Clostridium</i> , <i>Bacteroides</i>	Transform bile acids	Modulates lipid metabolism, gut micro composition
Synthesis of vitamins ²⁰²⁻²⁰³	Produces essential vitamins like K and B	Various metabolic pathways	<i>Bacteroides</i> , <i>Blifidobacterium</i>	Vitamin biosynthesis	Supports coagulation, energy, and neural he
Amino acid metabolism ²⁰⁴⁻²⁰⁵	Produces bioactive amines and neurotransmitter precursors	Decarboxylation and deamination pathways	<i>Enterococcus</i> , <i>Clostridium</i>	Metabolize amino acids	Influences gut-brain a mood regulation
Choline metabolism ²⁰⁶⁻²⁰⁷	Influences cardiovascular health through metabolite production	Conversion to TMA and TMAO	<i>Desulfovibrio</i> , <i>Bacteroides</i>	Convert choline to TMA	Affects cardiovascular
Polyphenol metabolism ¹⁻³⁰⁸	Produces antioxidant and anti-inflammatory compounds	Antioxidant response elements	<i>Lactobacillus</i> , <i>Blifidobacterium</i>	Metabolize polyphenols	Reduces oxidative stn inflammation
Indole production ^{209,210}	Impacts immune function and gut barrier integrity	Aryl hydrocarbon receptor	<i>Escherichia</i> , <i>Bacteroides</i>	Convert tryptophan to indole	Modulates immunity a gut barrier function
Methanogenesis ²¹¹⁻²¹²	Affects gut motility and digestion efficiency	Methanogenesis pathways	<i>Methanobrevibacter</i>	Convert CO ₂ and hydrogen to methane	Modulates gut motility gas production
Hydrogen sulfide production ²¹³⁻²¹⁴	Impacts mucosal integrity and inflammatory responses	Inflammatory signaling pathways	<i>Desulfovibrio</i> , <i>Blifidobacteria</i>	Produce hydrogen sulfide	Can increase inflammation, adverse affect gut health
Lactate utilization ^{215,216}	Converts lactate to SCFAs, contributing to energy supply	Metabolic fermentation pathways	<i>Bacteroides</i> , <i>Veillonella</i>	Convert lactate to SCFAs	Supports energy metabolism, gut health
Ethanolamine utilization ²¹⁷⁻²¹⁸	Influences cell signaling and lipid metabolism	Ethanolamine utilization pathways	<i>Parabacteroides</i> , <i>Escherichia</i>	Utilize ethanolamine	Modulates lipid metabolism, membrar integrity
Nitrate reduction ^{219,220}	Impacts nitrogen balance and nitric oxide pathways	Nitrate/nitrite reduction pathways	<i>Lactobacillus</i> , <i>Veillonella</i>	Reduce nitrates to nitrogen compounds	Modulates vascular function and microbial interactions
Xenobiotic metabolism ²²¹⁻²²²	Modifies metabolism of dietary and pharmaceutical compounds	Cytochrome P450 pathways	<i>Bacteroides</i> , <i>Escherichia</i>	Transform xenobiotics	Influences drug metabolism and detoxification
Modulation of mucus layer ²²³⁻²²⁵	Influences gut barrier and microbial niche	Mucus degradation and re-synthesis pathways	<i>Akkermansia</i> , <i>Bacteroides</i>	Degrade mucins, produce mucin-derived metabolites	Enhances gut barrier function, microbial bal
Energy extraction from non-digestible carbohydrates ²²⁶⁻²²⁸	Provides additional caloric intake through fermentation	Fermentation pathways	<i>Bacteroides</i> , <i>Ruminococcus</i>	Ferment complex carbohydrates	Contributes to caloric intake, impacts weight management

SCFA short-chain fatty acid, GPR G-protein coupled receptor, FXR farnesoid X receptor, TGR5 Takeda G protein-coupled receptor 5, TMAO trimethylamine N-oxide, TMO trimethylamine oxide.

[Jyoti & Dey, Met. H. and D. 25]

Gut microbiota and higher order diseases: some examples



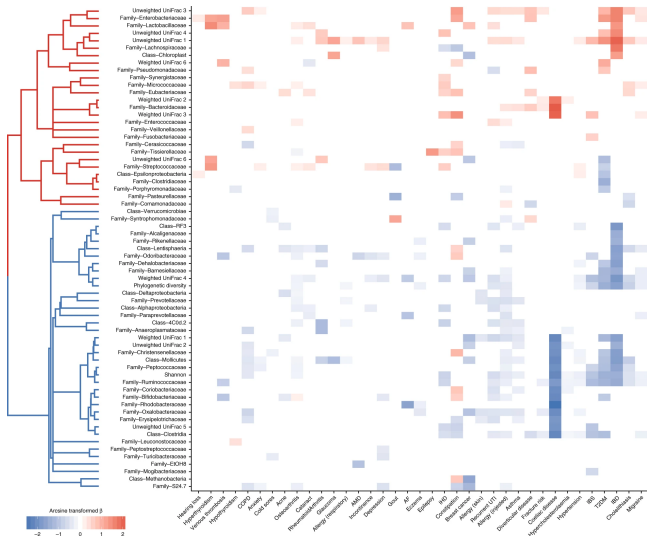
[O'Riordan et al. Cell Rep. Med. 25]

Gut microbiota and higher order diseases: some examples

Examples of known associations:

- Type II diabetes: 27 species variants identified as T2D signature [Mei et al Nature 24]
- Alzheimer's disease: signature and possible causal effects [Seo and Holtzman EMM 24]
- Parkinson's differential abundance of gut microbial species [Heintz-Buschart et al. *Mov. Disord.* 2018]

Gut microbiota and higher order diseases



[Jackson et al. Nature'18]

How to study microbial ecosystems?

- A *tiny* fraction of microbes are cultivable in a lab (probably less than few percent).
- Conducting biological/medical experiments is long and costly

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- Conducting biological/medical experiments is long and costly

How to study them, without observing them in the lab? How to study jointly humans and bacteria?

DNA: a universal way of coding (rather recent knowledge!)

Universal code

All known living organisms are coded through their DNA information. This determines to a large extent their morphologies and functions.

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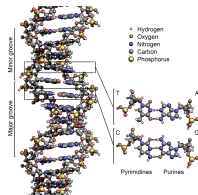
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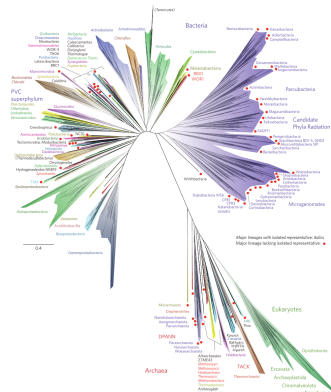
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- 1952 Hershey and Chase: DNA is known to encode genetic information
- 1953 Physical structure (double-helix) of DNA is solved using X-Ray diffraction by Franklin (but that's Watson & Crick who got the awards)



Origins and evolution of micro-organisms

Not a fixed knowledge: **we still continue to discover new branches of life:**



[Hug et al. 2016]

The Candidate Phyla Radiation (top right, in purple) has been discovered in 2016!

Origins and evolution of micro-organisms

Not a fixed knowledge: **we still continue to discover new branches of life:**

Resource | [Open access](#) | Published: 24 July 2025

Genome-resolved long-read sequencing expands known microbial diversity across terrestrial habitats

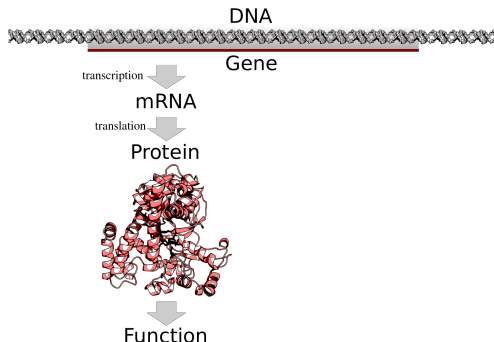
[Mantas Sereika](#), [Aaron James Mussig](#), [Chenjing Jiang](#), [Kalinka Sand Knudsen](#), [Thomas Bygh Nymann Jensen](#), [Francesca Petriglieri](#), [Yu Yang](#), [Vibeke Rudkjøbing Jørgensen](#), [Francesco Delogu](#), [Emil Aarre Sørensen](#), [Per Halkjær Nielsen](#), [Caitlin Margaret Singleton](#), [Philip Hugenholtz](#) & [Mads Albertsen](#) 

Nature Microbiology **10**, 2018–2030 (2025) | [Cite this article](#)

[Sereika et al. Nat. Microbio. 25]

How DNA determines an organism?

The big picture (for computer scientists): see video.



Proteins are responsible for most of the biological functions in organisms (biochemical reactions (enzymes), nutrient transportation, structural proteins, etc.)

Sequence-structure-function paradigm

Studying biological function through DNA information

From an organism to its **genome**...



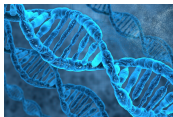
Organism

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Organism



DNA

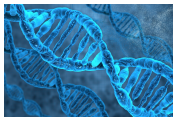


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Organism



DNA



Illumina/Nanopore

Studying biological function through DNA information

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Organism



DNA



Illumina/Nanopore



100's bp - few kbp

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Organism



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Illumina/Nanopore



Genomes

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Illumina/Nanopore



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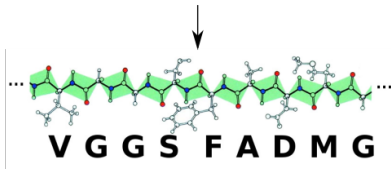
How does it help?

Bioinformatics: from genome to function

ACGATGTATTCAGCGATTACGATAAAGCTACGTAGTGGCA

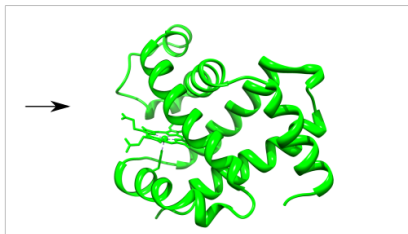
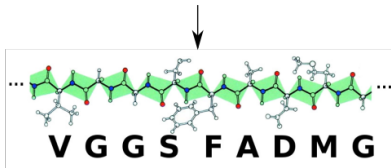
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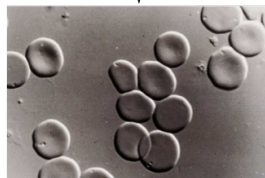
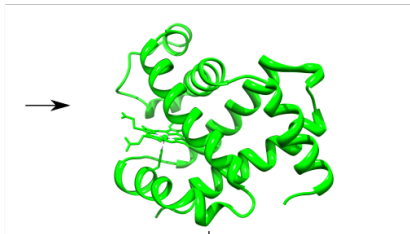
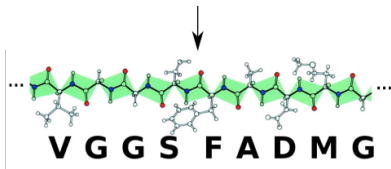
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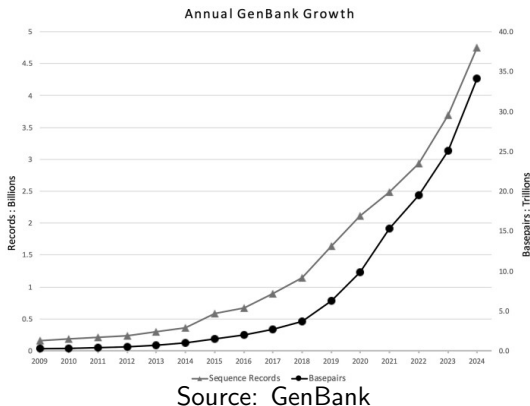


O₂ transport

What to do with these DNA sequences?

Exploding quantity of data

1977: first DNA sequencer.



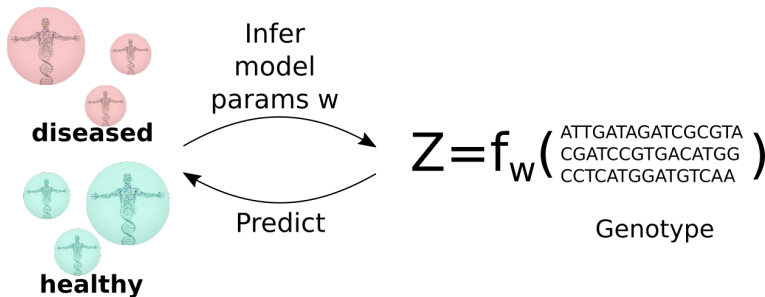
You can now sequence a human cell for less than a thousand euros.

Scientific questions call for various level of study

- *Genetics*: at gene level
- *Genomics*: at genome level
- *Metagenomics*: at whole community (several organisms) (partial) genomes level

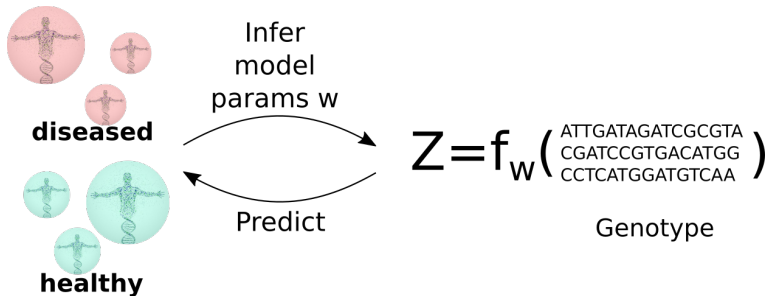
Example: association studies

Relates the variation of the genome to the phenotype. Recent review [Uffelmann et al. 2023]



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Define a predictor $f : \{A, T, C, G\}^M \rightarrow [0, 1]$ such that it minimizes a *loss* on a training set $(\vec{x}_1, z_1), \dots, (\vec{x}_N, z_N)$:

$$\min_f - \sum_{i=1}^N z_i \cdot \log f(\vec{x}_i) + (1 - z_i) \cdot \log(1 - f(\vec{x}_i))$$

Lots of data is not just for fun!

Human genome: what size? 1 Kbp? 10 Mbp?

Lots of data is not just for fun!

Human genome: 3Gbp.

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In practice, one “reference genome” and “only” $\approx 88 \cdot 10^6$ possible mutation places [The 1000 Genomes Project Consortium, 2015] .

Which mutation is responsible for a specific disease?

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Which mutation is responsible for a specific disease?
Better with **more** data... and fine statistics.

Not all is determined by the genome

Genome sequence is fully responsible for the hereditary phenotype.

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Epigenetics modifications (such as methylation) can be transferred to the progeny.

Scope of applications for DNA sequence data

Computationally processing DNA sequences has a huge number of applications, in particular in the fields of:

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-  clinics
- ecology 
-  biochemistry

...and sometimes involving the three at the same time!

Biology from the data perspective: not only DNA!

Biology brings various types of data, to get insights on various questions:

- Sequences **ATTCAGTACAT**
 - (Meta)Genomic: DNA sequence of one (several) organism

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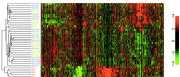
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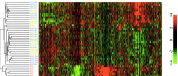
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- Images 
 - Neuroimaging
 - Cell imaging

- Mass spectrometry

- ...

Two reasons pushing computational biology forward

Computational biology

Biological data coupled with statistical models, machine learning, data visualization.

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Computational biology

Biological data coupled with statistical models, machine learning, data visualization.

- Availability of data
- Computing capacities

Artificial intelligence, second and recent breakthrough

Since 2021

Protein structure predictions (from DNA sequence) reached an accuracy equal to X-Ray crystallography using deep neural networks.

So now, using easy accessible DNA information, biologists can predict how protein interact with other compounds, and get more insight of the functions of the genes.

Hands-on

! ALERT !

Salmonella

OUTBREAK



Breaking news

Bad infections kill many people. Antibiotics do nothing.

Hands-on in two parts, you will develop tools to:

- Identify responsible gene
- Model the 3D structure of the involved protein

Disclaimer

- **No fully guided syllabus**
- Act as a junior professional
 - Analyze provided information, think of a solution
 - Ask/discuss with your colleagues
 - Ask/discuss with your senior colleague (me)

Plan of this session

First, think and plan - 1h.

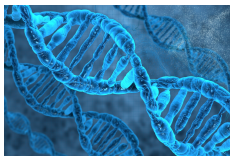
- Skim the context in the `hands-on/session1` on the git (15 min)
- Try to understand individually the work you will have to do and write down questions you have (10 min)
- Share your understanding with people in your group (10 min)
- We share together our understanding and elaborate a common strategy (20 min).

Then start developing - till the end :).

- Start developing T1, paying attention to pitfalls (noise in the data in particular)
- Build your own tests. You can make use of the data in the `hands-on/reference-data` directory

Genomics

From DNA to **reads**...



DNA



Illumina



reads ($\sim 250\text{bp}$)

$$\eta_{\text{err}} \approx 1\%$$

Genomics

From DNA to **reads**...



DNA



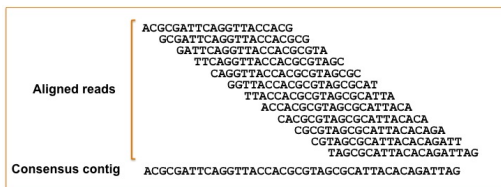
Illumina



reads ($\sim 250\text{bp}$)

$$\eta_{\text{err}} \approx 1\%$$

Assembly: from reads to **contigs**:



Sequencing data

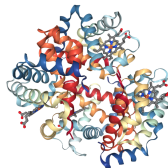
Two leading technologies:

- Illumina: pieces of sequences (called **reads**, 150-250bp)
 - +: reliable, about 1% sequencing errors.
 - -: short reads, only have local view of the genome
 - Errors: rare (1 over 100 bases) almost uniformly distributed, almost all **substitutions**.
- Nanopore: long reads, 10kb-100kb
 - +: long reads, easy to assemble, cheap and portable
 - -: high error rate
 - Errors: mostly insertion-deletion, mostly homopolymers (e.g. AAAAA)

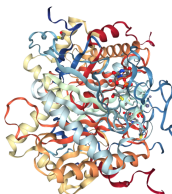


Predict the structure from sequence: the data

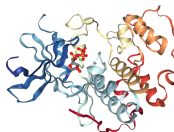
```
>1A3N:A|PDBID|CHAIN|SEQUENCE
VLSPADKTRNVKAAMGKVGGAHAGEYGAELER
MFLSPFTTKTYFFHPFDLSHGSAQVKGHGKKV
ADALTNVAIHVDMPNALGALSDLHAHKLRV
DPVNFKLLSHCLLVTLAAHLPAEFTPAVHAS
LDKFLASVSTVLTSKYR
```



```
>1HXP:A|PDBID|CHAIN|SEQUENCE
MTQFNVPVDHPHRRYNPLTGQWILVSPHRAKRPW
EGAQETPAKQVLPAHDPCFLCAGNVRVTGDKN
PDYTGTYVPTNDPAALMSDTPDAPESHDPLMRC
QSARGTSRVICFSPDHSKTLPELSVAALTEIVK
TWQEGTAEILGKTYPMVQVFENKGAAMGCSNPMP
HQIWMANSFLPNEAEREDRLQKEYFAHQKSPML
VDYVQRELADGSRVTVEIHLAVVPVWAANPF
ETLLLPKAHVLRITDLDQQRSDLAALAKKLTLS
RYDNLFCQCSFFPYSMGWHGAPFNGEENQHWQLHA
HFYFPLLRSATVRKFMVGYEMLAETQRDLTAEQ
AAERLRAVSDIHPRESGV
```



```
>1HCK:A|PDBID|CHAIN|SEQUENCE
MENPQKVEKIGEGTYGVVYKARNKLTGEVVAL
KKIRLDTETEGVGPSTAIRESLLKELNHPNIV
KLGDVIHTENKLYLVFEFLHQDLKKFMDASAL
TGIPFLPLIKSYLQLLQGLAFCHSHRVLRDL
KPNQLNINTEGAIKLADPGLARFGVVPRTYT
HEVVTLMYRAPEILLGCKYSTAVDIWSLGC
FAEMVTRRALFPDSEIDQLFRIFRTLGTPE
VWVPGVTSMPDYKPSFPKWARQDFSKVPPLD
EDGRSLLSQMLHYDPNKRISAKAALAHPPFQD
VTKPVPHLRL
```



CASP competition

Blind competition. Simple principle:

- a sequence is given
- have to predict the structure.

CASP competition

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13th CASP...

... AI wins !

Google's DeepMind

